

e-Tender Management System

Software Requirements Specification (SRS) & Project Documentation

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1. Introduction

1.1 Purpose

This document defines the requirements for the **e-Tender Management System**, a web application that digitises the full procurement cycle: publishing a notice and tender, receiving vendor bids, multi-level internal approvals (Technical Committee → Finance Committee → CFO), generating a three-vendor quotation comparison, and automatically informing the selected vendor.

1.2 Scope

The system covers:

- Notice and tender publication by the Purchase Officer.
- Vendor registration, login and bid submission through a vendor portal.
- Open tender (all vendors) and limited/invited tender (selected vendors only).
- Bid deadline control and manual tender opening.
- Technical Committee, Finance Committee and CFO approval workflow with optional supporting documents.
- Automatic generation of a three-vendor comparative quotation for validation.
- Auto-generated award email to the selected vendor.
- Audit trail, notifications and reports.

Out of scope for v1 (can be added later): payment processing, contract lifecycle management, online reverse auction, ERP/SAP integration, e-signature via government DSC.

1.3 Intended Audience

Product owner, business stakeholders, developers, QA engineers, UI/UX designers, DevOps and auditors.

2. Overall Description

2.1 Product Perspective

A standalone, role-based web application with two portals:

1. **Internal Portal** – Purchase Officer, Technical Committee, Finance Committee, CFO, Admin.
2. **Vendor Portal** – external vendors.

2.2 User Classes

Role	Description
Admin	Manages users, roles, committees, master data, system settings.
Purchase Officer (PO)	Creates notices/tenders, invites vendors, opens tender, forwards for approvals, sends award mail.
Technical Committee (TC) Member	Reviews technical compliance of bids and approves/rejects.
Finance Committee (FC) Member	Reviews commercial/financial aspects and approves/rejects.
CFO	Final approver of the selected quotation.
Vendor	Registers, logs in, views eligible tenders, submits/edits bids before deadline.

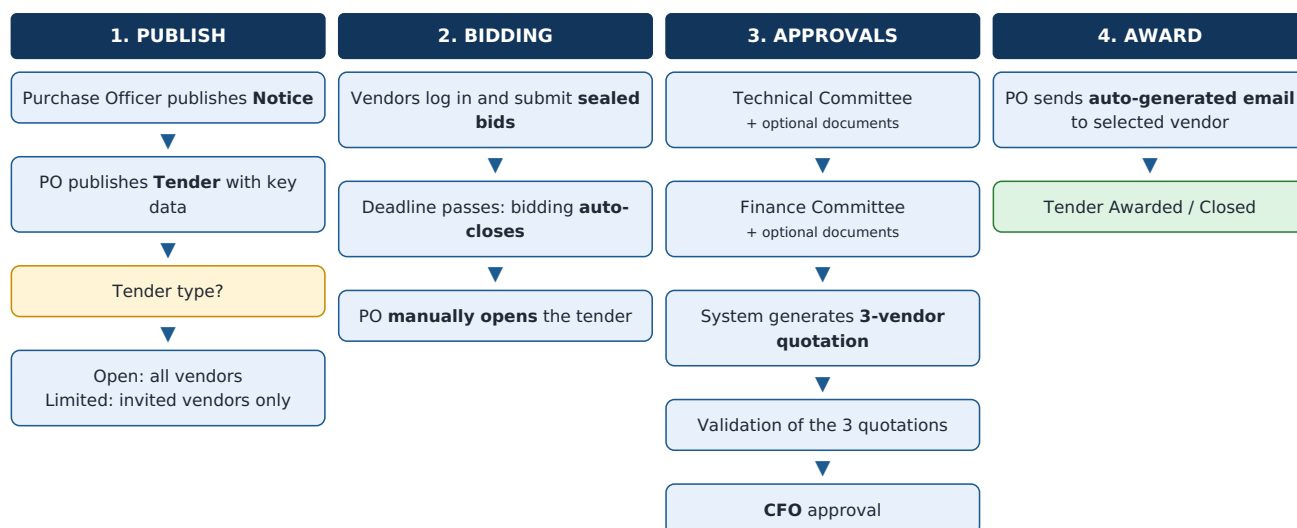
2.3 Operating Environment

Modern browsers (Chrome, Edge, Firefox, Safari), responsive for tablet/mobile. Cloud or on-premise Linux server deployment.

2.4 Constraints

- Bids must be sealed (not viewable by anyone) until the PO opens the tender after the deadline.
 - All approvals must be traceable (who, when, decision, remarks).
 - Late bids must be rejected by the system.
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3. Business Workflow



Lanes are read left to right. A rejection at Technical, Finance or CFO stage returns the tender to the Purchase Officer, who can re-evaluate, re-tender or cancel.

3.1 Step-by-Step Description

Step 1 - Notice and Tender Publication (Purchase Officer)

1. PO first publishes a **Notice** (pre-tender announcement).
2. PO then publishes the **Tender** with important data: title, tender number, description, category, quantity/specification, estimated value, EMD/security deposit (if any), eligibility criteria, terms & conditions, bid start date/time, bid end date/time, opening date, contact person, attachments.

Step 2 - Vendor Participation

1. Vendor logs into their portal.
2. Vendor sees tenders available to them (open tenders and tenders they were invited to).
3. Vendor submits a **bid** (price, delivery timeline, technical response, documents). Vendor may edit or withdraw until the deadline.
4. PO chooses tender visibility at creation: **Open to all vendors** or **Invite specific vendors**.

Step 3 - Bid Closure and Opening

1. After the configured deadline the system automatically closes bid submission.
2. PO manually **opens** the tender (bids become visible). Opening is logged.

Step 4 - Technical Committee Approval PO forwards the opened tender to the TC with optional supporting documents. TC members review and approve/reject with remarks.

Step 5 - Finance Committee Approval On TC approval, the tender goes to the FC (PO may add optional supporting documents). FC approves/rejects with remarks.

Step 6 - Three-Vendor Quotation Generation and Validation After FC approval, the system generates a **comparative statement of three vendor quotations** (default: three lowest technically-qualified priced bids, L1/L2/L3) for validation.

Step 7 - CFO Approval The validated three-vendor quotation goes to the CFO, who approves one vendor (normally L1, with justification required otherwise) or rejects.

Step 8 - Award Communication PO triggers (one click) the **auto-generated email** to the selected vendor. Optionally, regret emails go to the other bidders.

4. User Roles & Permissions

Feature	Admin	PO	TC	FC	CFO	Vendor
Manage users/roles/committees	✓					
Create/publish notice		✓				
Create/publish tender		✓				
Invite vendors		✓				
View tender (public/invited)	✓	✓	✓*	✓*	✓*	✓
Submit / edit / withdraw bid						✓
View bids before opening	-	-	-	-	-	Own only
Open tender		✓				
Forward to TC/FC		✓				
Technical approve/reject			✓			
Financial approve/reject				✓		
Validate 3-quotation statement		✓		✓		
Final approval					✓	
Send award email		✓				
View audit log / reports	✓	✓	limited	limited	✓	-

* Only when the tender is assigned to them in the workflow.

5. Functional Requirements

Priority: **M** = Must, **S** = Should, **C** = Could.

5.1 Authentication & User Management (AUTH)

ID	Requirement	Pri
AUTH-01	Users log in with email/username and password; vendors and internal users use separate portals/entry points.	M
AUTH-02	Passwords stored hashed (bcrypt/argon2); password policy enforced.	M
AUTH-03	Two-factor authentication (OTP via email/SMS or authenticator) for internal users and optionally for vendors.	S
AUTH-04	Role-based access control (RBAC) enforced on every page and API.	M
AUTH-05	Account lockout after repeated failed login; forgot/reset password by email link.	M
AUTH-06	Session timeout after inactivity (configurable, default 20 min).	M
AUTH-07	Admin can create users, assign roles, activate/deactivate accounts.	M
AUTH-08	Admin can define Technical and Finance committees and their members.	M

5.2 Vendor Management (VEN)

ID	Requirement	Pri
VEN-01	Vendor self-registration with company name, contact, email, phone, address, GST/Tax ID, PAN, bank details (optional), category of supply.	M
VEN-02	Email verification and admin/PO approval of vendor registration before the vendor can bid.	M
VEN-03	Vendor uploads compliance documents (registration certificate, tax certificate, experience, etc.) with expiry dates.	S
VEN-04	Vendor can be blacklisted/suspended; blacklisted vendors cannot see or bid on tenders.	M
VEN-05	Vendors are categorised (e.g., IT, civil, stationery) to enable targeted invitations.	S

VEN-06	Vendor profile edit with change history.	S
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5.3 Notice Management (NOT)

ID	Requirement	Pri
NOT-01	PO can create a notice: title, notice number (auto), description, publish date, attachments.	M
NOT-02	Notice states: Draft → Published → Archived.	M
NOT-03	A tender can only be published if it is linked to a published notice (configurable rule).	M
NOT-04	Published notices are visible to all registered vendors (and optionally on a public page).	M
NOT-05	Notification to vendors when a notice is published.	S

5.4 Tender Management (TEN)

ID	Requirement	Pri
TEN-01	PO creates a tender with: tender number (auto), title, linked notice, category, description/specification, quantity & unit, estimated value, currency, EMD/security amount, eligibility criteria, payment & delivery terms, evaluation criteria, contact person.	M
TEN-02	PO sets bid start date-time , bid end date-time (deadline) and planned opening date-time .	M
TEN-03	PO attaches documents (BOQ, specifications, drawings, T&C) – PDF/DOCX/XLSX/images with size and type limits.	M
TEN-04	PO chooses visibility : (a) <i>Open</i> – all approved vendors, or (b) <i>Limited/Invited</i> – only selected vendors.	M
TEN-05	For limited tenders, PO selects vendors from the vendor list (search/filter by category); only invited vendors can see and bid.	M
TEN-06	PO can save tender as Draft, edit, then Publish. After publishing, critical fields can change only through a corrigendum/amendment that notifies all vendors.	M
TEN-07	PO can extend the deadline (with reason, logged, vendors notified) or cancel a tender (with reason).	S
TEN-08	Vendors can raise clarification questions before the deadline; PO can reply publicly to all bidders (Q&A).	C
TEN-09	Tender clone/duplicate feature.	C
TEN-10	Tender list with search, filter (status, date, category) and export.	M

5.5 Vendor Bidding (BID)

ID	Requirement	Pri
BID-01	Vendor dashboard lists tenders: Open, Invited, Submitted, Closed, Awarded.	M
BID-02	Vendor views tender details and downloads tender documents.	M
BID-03	Vendor submits a bid containing: quoted price (total and per-item if BOQ), taxes, delivery period, validity of offer, technical response/remarks, and uploaded documents.	M
BID-04	System accepts bids only between bid start and bid end times (server time). Late submission is blocked.	M
BID-05	One active bid per vendor per tender; vendor may edit or withdraw until the deadline; every revision is versioned.	M
BID-06	Bid content is encrypted/sealed ; nobody (including PO and Admin) can view bid contents before tender opening.	M
BID-07	Vendor receives a submission acknowledgement (on screen + email) with a receipt number and timestamp.	M
BID-08	Vendor cannot see other vendors' bids at any time.	M
BID-09	Countdown timer to deadline shown to vendors.	S
BID-10	EMD/security deposit proof upload (if required by tender).	S

5.6 Tender Opening (OPN)

ID	Requirement	Pri
OPN-01	After the deadline, bid submission closes automatically (scheduler).	M

OPN-02	The Open Tender action is available only to the PO, only after the deadline, and is performed manually .	M
OPN-03	On opening, bids are decrypted and visible to authorised internal users; opening event (user, timestamp, IP) is logged.	M
OPN-04	System displays a bid opening summary: number of bids, vendor names, quoted price, document list.	M
OPN-05	Tender with zero bids can be re-published or cancelled.	S
OPN-06	Optional two-person rule for opening (PO + witness) – configurable.	C

5.7 Approval Workflow (APR)

5.7.1 Technical Committee

ID	Requirement	Pri
APR-01	PO forwards the opened tender to the Technical Committee with remarks and optional supporting documents .	M
APR-02	TC members receive notification and can review tender, all bids and documents.	M
APR-03	TC records per-bid technical evaluation (Qualified / Not qualified, score optional, remarks).	M
APR-04	TC decision: Approve, Reject, or Send back for clarification . Remarks mandatory on reject/send-back.	M
APR-05	Approval mode configurable: any one member, majority, or all members must approve.	S

5.7.2 Finance Committee

ID	Requirement	Pri
APR-10	Only after TC approval, PO forwards to the Finance Committee with optional supporting documents .	M
APR-11	FC reviews technically-qualified bids, budget availability and commercial terms.	M
APR-12	FC decision: Approve, Reject, or Send back . Remarks mandatory on reject/send-back.	M

5.7.3 Three-Vendor Quotation Generation & Validation

ID	Requirement	Pri
APR-20	After FC approval, the system automatically generates a comparative statement of three vendor quotations (default rule: the three lowest-priced, technically-qualified bids = L1, L2, L3).	M
APR-21	If fewer than three qualified bids exist, system warns and allows proceeding with justification (configurable rule).	M
APR-22	Statement shows vendor, price, taxes, delivery period, validity, technical score, deviations from L1.	M
APR-23	Statement is validated (checked and confirmed) by the PO and/or FC before it goes forward; validator name and time are recorded.	M
APR-24	Comparative statement exportable to PDF/Excel.	S

5.7.4 CFO Approval

ID	Requirement	Pri
APR-30	Validated statement is routed to the CFO.	M
APR-31	CFO selects one vendor to award (default L1). If a non-L1 vendor is chosen, a justification is mandatory.	M
APR-32	CFO decision: Approve, Reject, or Send back .	M
APR-33	Delegation: CFO can delegate approval to another user for a set period (with audit).	C

5.7.5 General Approval Rules

ID	Requirement	Pri
APR-40	Stages are strictly sequential: PO → TC → FC → Quotation validation → CFO → Award mail. No stage can be skipped.	M
APR-41	Every decision stores approver, decision, remarks, timestamp, and IP.	M
APR-42	A rejected tender returns to PO with reasons; PO may re-evaluate, re-tender or cancel.	M

APR-43	Approval SLA reminders/escalation if a stage is pending beyond configured days.	S
APR-44	Supporting documents at TC/FC stages are optional; system supports multiple files each.	M

5.8 Award & Auto Email (AWD)

ID	Requirement	Pri
AWD-01	After CFO approval the PO sees a Send Award Email action.	M
AWD-02	System auto-generates the email using a template populated with tender number, title, awarded value, delivery terms, next steps, contact details and (optionally) a PDF award letter.	M
AWD-03	Email is sent to the selected vendor's registered email; PO can preview before sending.	M
AWD-04	Send status (Sent/Failed), timestamp and message ID stored; failed emails can be retried.	M
AWD-05	Optional regret/thank-you emails to non-selected bidders.	S
AWD-06	Vendor can view the award status in the portal and acknowledge/accept the award.	S
AWD-07	Tender status becomes Awarded after the mail is sent, then can be Closed .	M

5.9 Notifications (NTF)

ID	Requirement	Pri
NTF-01	In-app and email notifications for each significant event (see section 9).	M
NTF-02	Users can view notification history.	S
NTF-03	SMS/WhatsApp notifications.	C

5.10 Reports & Dashboard (RPT)

ID	Requirement	Pri
RPT-01	PO dashboard: tenders by status, upcoming deadlines, pending actions.	M
RPT-02	Approver dashboard: pending approvals, aging.	M
RPT-03	Reports: tender register, vendor participation, bid comparison, approval turnaround time, awarded value, savings vs. estimate.	S
RPT-04	Export to Excel/PDF.	S

5.11 Audit Log (AUD)

ID	Requirement	Pri
AUD-01	Immutable log of all critical actions (login, publish, bid submit/edit/withdraw, open, approvals, email send, config changes).	M
AUD-02	Admin/auditor can search and export logs.	M

6. Tender & Bid Status Lifecycle

6.1 Tender Status

Status	Meaning	Next allowed
DRAFT	Being prepared by PO	PUBLISHED
PUBLISHED	Live; bidding window may not have started	BIDDING_OPEN
BIDDING_OPEN	Vendors can submit bids	BIDDING_CLOSED
BIDDING_CLOSED	Deadline passed; bids sealed	OPENED
OPENED	PO opened tender; bids visible	TECHNICAL_REVIEW
TECHNICAL_REVIEW	With Technical Committee	FINANCE_REVIEW / REJECTED / SENT_BACK

FINANCE_REVIEW	With Finance Committee	QUOTATION_VALIDATION / REJECTED / SENT_BACK
QUOTATION_VALIDATION	3-vendor statement generated, being validated	CFO_REVIEW
CFO_REVIEW	With CFO	APPROVED / REJECTED / SENT_BACK
APPROVED	CFO approved; awaiting award mail	AWARDED
AWARDED	Award email sent	CLOSED
CLOSED	Completed	-
REJECTED / CANCELLED	Terminated with reason	-

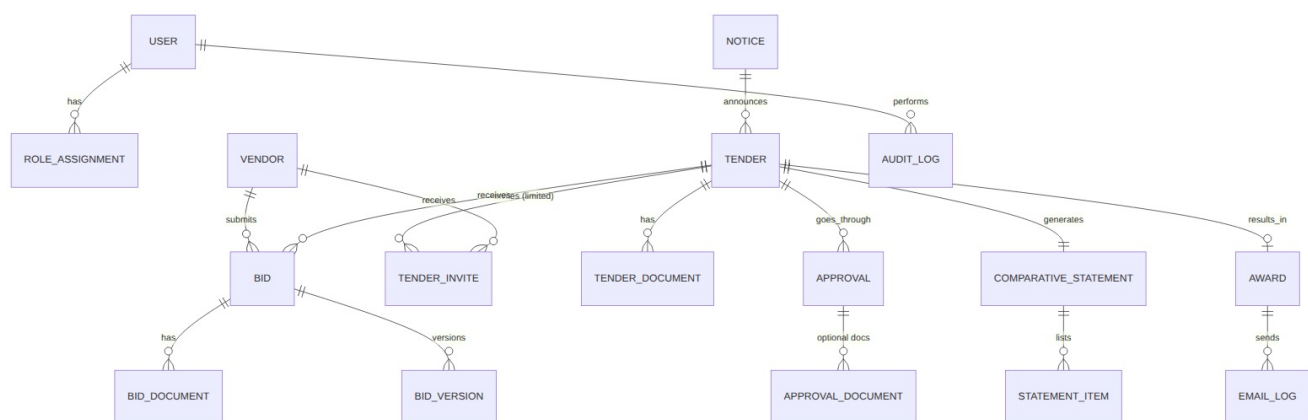
6.2 Bid Status

DRAFT → SUBMITTED → (REVISED) → WITHDRAWN | OPENED → TECH_QUALIFIED / TECH_DISQUALIFIED → SHORTLISTED (L1-L3) → SELECTED / NOT_SELECTED

7. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Pages load in ≤ 3 s under normal load; support ≥ 500 concurrent vendors, especially near deadlines.
NFR-02	Availability	99.5%+ uptime; higher tolerance near bid deadlines (avoid maintenance windows).
NFR-03	Security	HTTPS/TLS 1.2+, OWASP Top 10 protections, encryption at rest for bids and documents.
NFR-04	Time integrity	All deadlines enforced on server time (NTP-synced) ; timezone stored in UTC, displayed in local time.
NFR-05	Scalability	Horizontally scalable stateless application tier; async queue for emails and file processing.
NFR-06	Usability	Responsive UI, clear status indicators, accessibility (WCAG 2.1 AA target).
NFR-07	Data retention	Tender records, bids and audit logs kept for a configurable period (e.g., 7 years).
NFR-08	Backup/DR	Daily backups, RPO ≤ 24 h (lower preferred), RTO ≤ 4 h.
NFR-09	File handling	Max file size configurable (e.g., 25 MB), antivirus scan on upload, allowed types whitelist.
NFR-10	Maintainability	Modular code, automated tests ($\geq 70\%$ coverage on business logic), CI/CD pipeline.
NFR-11	Auditability	All state changes traceable; logs tamper-evident.
NFR-12	Localization	English initially; architecture ready for multi-language and multi-currency.

8. Data Model (Logical)



8.1 Main Entities

Entity	Key fields
User	id, name, email, password_hash, role(s), status, last_login, 2FA settings

Vendor	id, company_name, contact_name, email, phone, address, tax_id, category, status (Pending/Approved/Suspended/Blacklisted), documents
Notice	id, notice_no, title, description, status, published_by, published_at, attachments
Tender	id, tender_no, notice_id, title, description, category, quantity, unit, estimated_value, currency, emd_amount, eligibility, terms, visibility (OPEN/LIMITED), bid_start_at, bid_end_at, opening_at, status, created_by, opened_by, opened_at
TenderInvite	id, tender_id, vendor_id, invited_at, viewed_at
TenderDocument	id, tender_id, file_name, file_path, type, uploaded_by
Bid	id, tender_id, vendor_id, status, current_version, receipt_no, submitted_at
BidVersion	id, bid_id, version_no, encrypted_payload (price, taxes, delivery days, validity, remarks), created_at
BidDocument	id, bid_version_id, file_name, file_path
Approval	id, tender_id, stage (TECHNICAL/FINANCE/CFO), forwarded_by, forwarded_at, approver_id, decision, remarks, decided_at
ApprovalDocument	id, approval_id, file_name, file_path (optional)
TechnicalEvaluation	id, tender_id, bid_id, result, score, remarks, evaluated_by
ComparativeStatement	id, tender_id, generated_at, validated_by, validated_at, status
StatementItem	id, statement_id, bid_id, rank (L1/L2/L3), price, delivery_days, notes
Award	id, tender_id, bid_id, vendor_id, awarded_value, approved_by (CFO), justification, status
EmailLog	id, award_id/tender_id, recipient, subject, body, status, sent_at, message_id, error
Notification	id, user_id/vendor_id, type, message, read_at
AuditLog	id, actor_id, action, entity_type, entity_id, old_value, new_value, ip, timestamp
Config	key, value (e.g., approval mode, min bids, max file size)

9. Notifications & Email Templates

9.1 Event-Notification Matrix

Event	Recipients	Channel
Vendor registered	PO/Admin	In-app + email
Vendor approved/rejected	Vendor	Email
Notice published	All approved vendors	Email + in-app
Tender published (open)	All approved vendors	Email + in-app
Tender published (limited)	Invited vendors only	Email + in-app
Bid submitted / revised / withdrawn	Vendor (receipt)	Email + in-app
Deadline approaching (24 h, 1 h)	Vendors who haven't bid / draft bids	Email + in-app
Deadline extended / corrigendum	All eligible vendors	Email + in-app
Bidding closed	PO	In-app
Forwarded to TC / FC / CFO	Relevant approvers	Email + in-app
Approval / rejection / send-back	PO (+ next-stage approvers on approval)	Email + in-app
Comparative statement ready	PO / FC	In-app
Award email sent	Selected vendor (+ PO copy)	Email
Pending approval reminder	Approver	Email

9.2 Award Email Template (auto-generated)

Subject: Award of Tender – {{tender_no}} – {{tender_title}}

Dear {{vendor_contact_name}},

We are pleased to inform you that {{vendor_company_name}} has been selected for the tender detailed below, following evaluation and approval by our competent authority.

Tender No. : {{tender_no}}
Tender Title : {{tender_title}}
Awarded Value : {{currency}} {{awarded_value}}
Delivery Period : {{delivery_period}}
Offer Validity : {{offer_validity}}

Next steps:

1. Please confirm acceptance of this award by {{acceptance_date}} through the vendor portal or by replying to this email.
2. Our procurement team will contact you regarding the contract/purchase order and other formalities.

Contact: {{purchase_officer_name}}, {{purchase_officer_email}}, {{phone}}

Regards,
Procurement Department
{{organization_name}}

9.3 Regret Email Template (optional)

Subject: Update on Tender - {{tender_no}}

Dear {{vendor_contact_name}},

Thank you for participating in tender {{tender_no}} – {{tender_title}}.
After evaluation, your bid was not selected on this occasion. We appreciate your interest and look forward to your participation in future tenders.

Regards,
Procurement Department

10. Screens / UI List

10.1 Vendor Portal

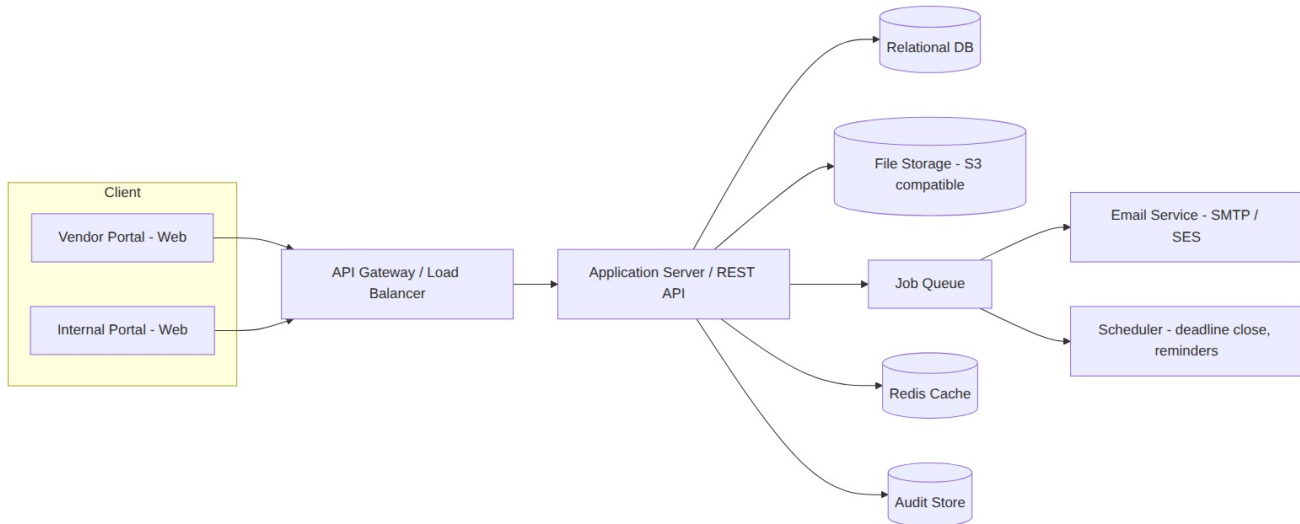
1. Login / Register / Forgot password
2. Dashboard (available tenders, invited tenders, my bids, notices, deadlines)
3. Notice list & detail
4. Tender list & detail (documents, dates, countdown)
5. Bid submission form (price, delivery, documents, review & confirm)
6. My bids (status, versions, receipt, edit/withdraw)
7. Award notifications
8. Profile & compliance documents

10.2 Internal Portal

1. Login (with 2FA)
2. Role-based dashboard
3. Notice create/list
4. Tender create (multi-step wizard: Details → Dates → Documents → Visibility/Invite vendors → Review → Publish)
5. Tender list & detail (timeline of workflow)

6. Bid opening screen (**Open Tender** button, enabled after deadline)
7. Forward to TC/FC screen (remarks + optional documents)
8. Committee review screen (bid comparison, evaluation form, approve/reject)
9. Comparative statement (L1/L2/L3) with validation
10. CFO approval screen
11. Award email preview & send
12. Vendor management
13. User/role/committee management (Admin)
14. Reports & audit log
15. System settings & email templates

11. Suggested System Architecture & Tech Stack



Layer	Suggested option
Frontend	React / Next.js (or Angular) + Tailwind/Material UI
Backend	Node.js (NestJS/Express) or Java Spring Boot or Python Django/FastAPI or Laravel
Database	PostgreSQL (or MySQL)
File storage	S3-compatible object storage with server-side encryption
Queue/Scheduler	BullMQ/Redis, RabbitMQ, or Celery
Email	SMTP / Amazon SES / SendGrid
Auth	JWT + refresh tokens, or OAuth2/OIDC; RBAC middleware
Hosting	Docker + Kubernetes / cloud VM; CI/CD via GitHub Actions
Monitoring	Prometheus/Grafana, Sentry, centralized logs

11.1 Key Design Decisions

- **Sealed bids:** bid payloads encrypted with an application key (or per-tender key) and only decrypted by the "Open Tender" action; access blocked at API level before opening.
- **Deadline enforcement:** server-side scheduler flips tender status to BIDDING_CLOSED; API also validates timestamps on each submit request.
- **Workflow engine:** a state-machine service enforces allowed transitions and role checks; each transition writes an audit record.
- **Idempotent email sending:** email jobs are queued, logged, retried on failure.

12. API Overview (REST, indicative)

Area	Endpoint	Method	Role
Auth	/auth/login, /auth/logout, /auth/refresh, /auth/forgot-password	POST	All
Vendors	/vendors (register), /vendors/{id}/approve	POST/PATCH	Vendor / PO
Notices	/notices, /notices/{id}/publish	POST/GET/PATCH	PO
Tenders	/tenders, /tenders/{id}, /tenders/{id}/publish	POST/GET/PUT/PATCH	PO
Invites	/tenders/{id}/invites	POST/GET/DELETE	PO
Corrigendum	/tenders/{id}/corrigendum, /tenders/{id}/extend	POST	PO
Bids	/tenders/{id}/bids (submit), /bids/{id} (edit), /bids/{id}/withdraw	POST/PUT/PATCH	Vendor
Opening	/tenders/{id}/open	POST	PO
Forwarding	/tenders/{id}/forward/technical, /forward/finance	POST	PO
Decisions	/tenders/{id}/technical-decision, /finance-decision, /cfo-decision	POST	TC / FC / CFO
Statement	/tenders/{id}/comparative-statement, /validate	GET/POST	PO / FC
Award	/tenders/{id}/award/send-email	POST	PO
Files	/files/upload, /files/{id}	POST/GET	Authorised
Reports	/reports/*	GET	Internal
Audit	/audit-logs	GET	Admin

13. Security, Audit & Compliance

1. **Access control:** RBAC + tender-level permission checks (a vendor sees only open tenders and tenders they're invited to).
2. **Data protection:** encryption in transit (TLS) and at rest; secrets in a vault; no sensitive data in logs.
3. **Application security:** input validation, parameterised queries, CSRF/XSS protection, rate limiting, secure file upload with AV scan.
4. **Segregation of duties:** the user who creates the tender cannot approve it; an approver cannot approve at more than one stage for the same tender (configurable).
5. **Audit trail:** append-only, includes IP and user agent; exportable for audits.
6. **Conflict-of-interest declaration:** committee members declare no conflict before reviewing (optional).
7. **Privacy:** vendor data handled according to applicable data-protection law; retention and deletion policy defined.
8. **Backups:** encrypted, tested restore process.

14. Test Scenarios / Acceptance Criteria

#	Scenario	Expected result
T1	PO publishes notice, then tender with all mandatory fields	Tender visible to eligible vendors; audit entry created
T2	PO publishes limited tender inviting 3 vendors	Only those 3 see it; others get "not found"
T3	Vendor submits bid before deadline	Receipt number + email issued
T4	Vendor attempts to bid after deadline	Blocked with clear message
T5	Vendor edits bid before deadline	New version stored; previous kept in history
T6	PO/Admin tries to view bids before opening	Access denied
T7	PO tries to open tender before deadline	Action disabled/denied
T8	PO opens tender after deadline	Bids visible to authorised users; event logged
T9	PO forwards to TC without supporting documents	Allowed (documents optional)
T10	TC rejects with no remarks	Validation error; remarks required

T11	TC approves → FC forward	FC receives; status updates; cannot skip TC
T12	FC approves	System generates comparative statement of 3 vendors (L1-L3)
T13	Fewer than 3 qualified bids	Warning + justification required to proceed
T14	Statement validated → CFO	CFO receives validated statement
T15	CFO picks non-L1 vendor without justification	Blocked
T16	CFO approves; PO sends award mail	Email sent to selected vendor only; status = Awarded; log stored
T17	Email send fails (SMTP down)	Status Failed, retry available, PO alerted
T18	Vendor from another tender tries direct URL access	Access denied (authorization check)
T19	Concurrent submissions near deadline	All valid bids accepted; none after cut-off
T20	Blacklisted vendor logs in	Cannot view/submit tenders

15. Assumptions, Open Questions & Risks

15.1 Assumptions (based on your description)

1. "Notice" is a pre-tender announcement; the tender is published after it and linked to it.
2. "Manually a time will be after that" means the PO sets a deadline; when it passes, bidding closes, and the PO **manually** opens the tender.
3. "Generate 3 vendor quotation for validation" means the system produces a comparative statement of the **top three** vendor quotations (default L1-L3 by price among technically-qualified bids), which is validated before the CFO.
4. The CFO chooses the final vendor from those three, and the award email goes **only to that vendor** (regret mails optional).
5. TC and FC are groups; a configurable rule decides how many members must approve.

15.2 Open Questions for Stakeholders

1. Should vendors be able to see the notice publicly (without login)?
2. How is the "top 3" chosen - lowest price only, or a weighted technical + financial score (QCBS)?
3. Should the tender opening require two people (PO + witness)?
4. Is the EMD / bid security to be handled inside the app (online payment) or just uploaded as proof?
5. What happens on rejection - re-tender, negotiation, or cancel? Should there be a "send back" loop?
6. Approval mode: one member, majority or all members for TC and FC?
7. Any approval value thresholds (e.g., CFO only above a certain amount)?
8. Are digital signatures required on approvals or the award letter?
9. Integration needs (ERP, accounting, SMS gateway)?
10. Expected number of tenders and vendors per year (for sizing)?
11. Languages and currencies required?

15.3 Risks & Mitigation

Risk	Mitigation
Bid leakage before opening	Encryption, strict API guards, audit, no bulk exports pre-opening
Server overload at deadline	Autoscaling, queueing uploads, load testing
Disputes about time	Server-time enforcement, submission receipts, NTP sync
Approval delays	SLA reminders, escalation, delegation
Email failures	Queue + retry + status dashboard
Scope creep	Phase-wise delivery, change-control process

16. Delivery Roadmap (Suggested)

Phase	Duration (approx.)	Deliverables
0. Discovery	1-2 weeks	Sign-off on SRS, clarify open questions, UI wireframes
1. Foundation	2-3 weeks	Auth, RBAC, user/vendor management, admin
2. Tender core	3-4 weeks	Notice, tender, invitations, documents, vendor portal listing
3. Bidding	3 weeks	Bid submission, versioning, sealing, deadline scheduler, opening
4. Approvals	3-4 weeks	TC/FC/CFO workflow, comparative statement, validation
5. Award & comms	2 weeks	Award email, templates, notifications, regret mails
6. Reports & hardening	2-3 weeks	Dashboards, reports, audit, security testing, performance tests
7. UAT & go-live	2 weeks	UAT, training, deployment, support

MVP suggestion: Phases 1-5 with minimum reports.

17. Glossary

Term	Meaning
Notice	Pre-tender public announcement of intent to procure
Tender	Formal invitation to vendors to submit priced bids
Open tender	Visible to all approved vendors
Limited/Invited tender	Visible only to selected vendors
Bid / Quotation	Vendor's offer including price and terms
Sealed bid	Bid content hidden until official opening
EMD	Earnest Money Deposit / bid security
TC	Technical Committee
FC	Finance Committee
CFO	Chief Financial Officer (final approver)
L1, L2, L3	Lowest, second-lowest, third-lowest priced qualified bids
Comparative Statement	Side-by-side comparison of shortlisted quotations
Corrigendum	Official amendment to a published tender
RBAC	Role-Based Access Control

End of document.